Time Series Forecasting of Oil Prices using ARIMA and LSTM

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GitHub Link : https://github.com/Lohitraj66/Time-Series-Modelling-Case-Study.git

1. Introduction

Forecasting oil prices is important for economic planning, financial markets, and energy policy. Oil prices exhibit strong temporal patterns and are influenced by market conditions, geopolitical events, and supply-demand dynamics. Because of this, time series forecasting methods are commonly used to model and predict oil price movements.

This report investigates the use of two different modelling approaches for forecasting daily oil prices:

- **ARIMA (AutoRegressive Integrated Moving Average)** – a classical statistical time series model.
- **LSTM (Long Short-Term Memory neural network)** – a deep learning approach designed for sequential data.

The objective is to compare these two approaches in terms of prediction accuracy and modelling capability. The dataset contains 500 daily oil price observations ranging from September 2024 to February 2026. The analysis includes exploratory data analysis, stationarity testing, model training, validation, and forecasting.

2. Exploratory Data Analysis

Before applying forecasting models, the data was explored to understand its structure and statistical properties.

2.1 Time Series Behaviour

The oil price time series was plotted to observe trends and fluctuations over time.

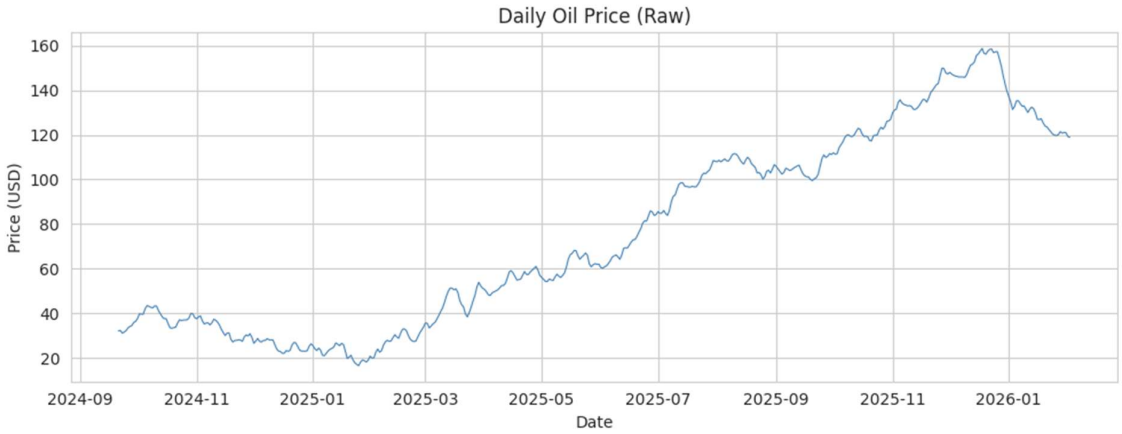

Figure 1: Raw oil price time series

The plot shows that oil prices fluctuate over time with noticeable short-term volatility. The presence of visible trends suggests that the series is not stationary, which is a key requirement for many time series models such as ARIMA.

2.2 Distribution of Prices

A histogram and box plot were used to analyse the distribution of oil prices.

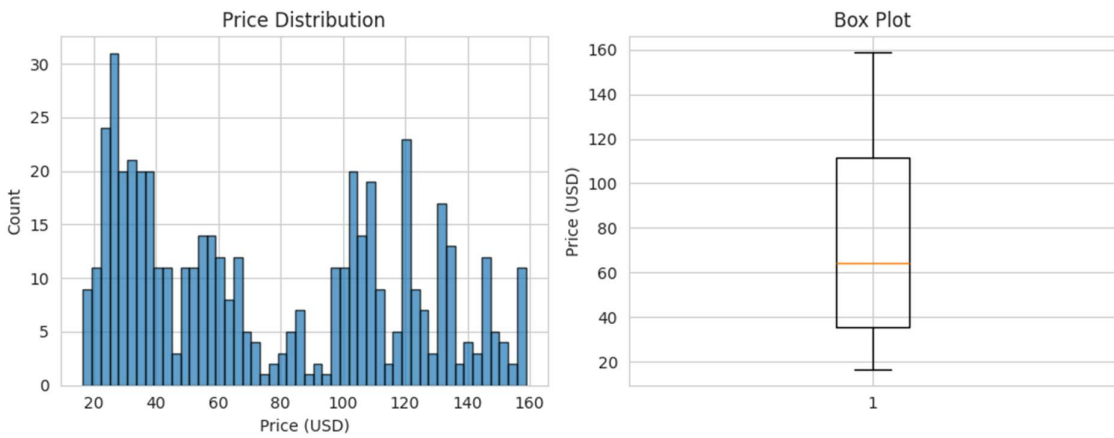

Figure 2: Histogram and box plot of oil prices

The histogram shows the overall spread of price values, while the box plot highlights potential outliers and the central tendency. The distribution appears slightly skewed, indicating that price fluctuations are not perfectly symmetric.

Understanding the distribution helps determine whether transformations or differencing may be required before modelling.

3. Stationarity Testing

Stationarity is a fundamental assumption for many time series models. A stationary series has constant mean and variance over time.

To test stationarity, the **Augmented Dickey-Fuller (ADF) test** was applied.

ADF Test on Original Series

ADF Statistic: -0.5462
p-value: 0.8827

Since the p-value is much greater than 0.05, we fail to reject the null hypothesis of a unit root. This indicates that the series is non-stationary.

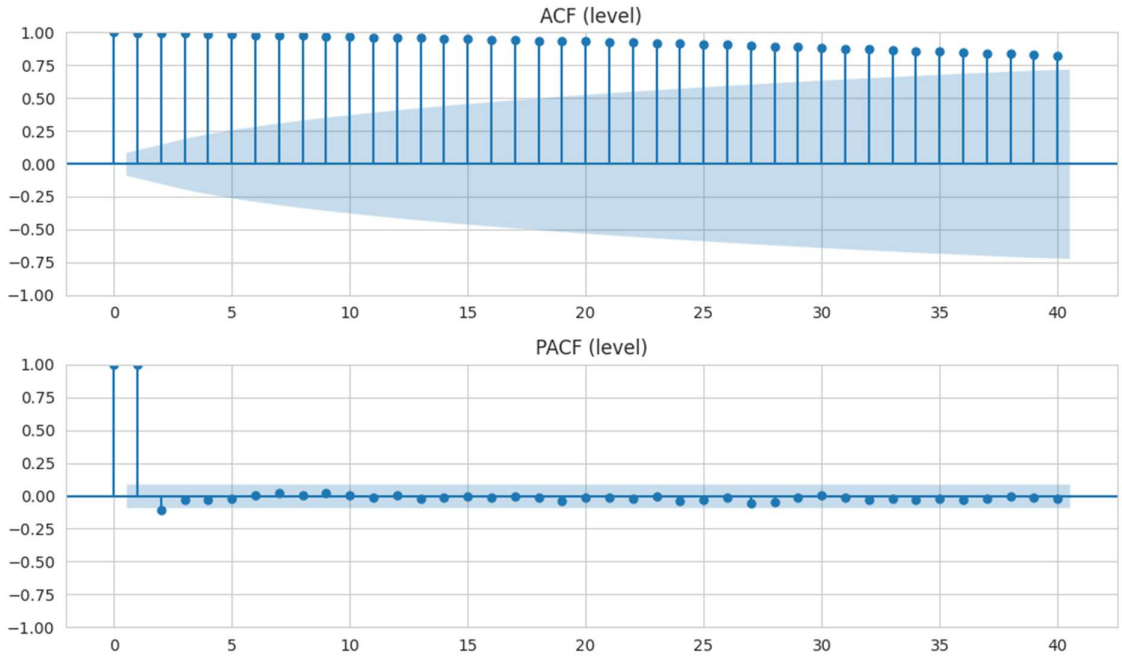

Figure 3: ACF and PACF plots

First Differencing

To remove non-stationarity, the first difference of the series was computed.

ADF Statistic (1st difference): -4.5415
p-value: 0.0002

The p-value is now significantly below 0.05, meaning the differenced series is stationary.

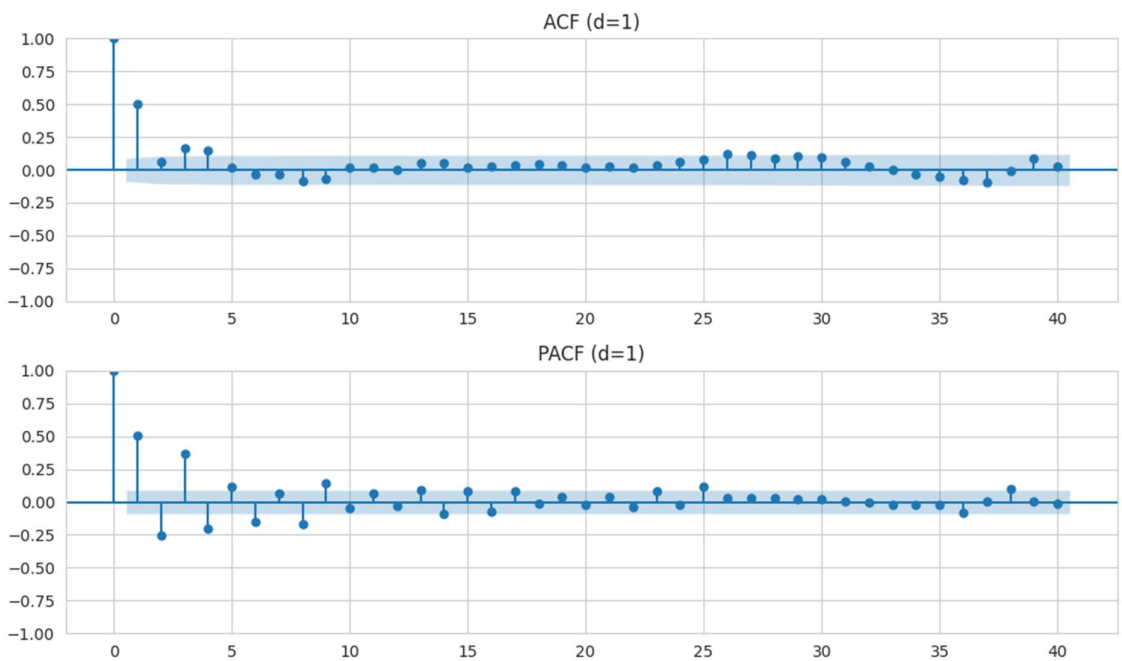

Figure 4: ACF and PACF plots for differenced series

The ACF and PACF plots help determine suitable ARIMA parameters by showing the correlation structure of the time series.

4. Methodology

4.1 Data Splitting

To evaluate the predictive performance of the models, the dataset was split into:

- Training set: 80% of data (400 observations)
- Validation set: 20% of data (100 observations)

The models were trained using the training data and evaluated using the validation data.

Model performance was assessed using two evaluation metrics:

- RMSE (Root Mean Squared Error) – penalizes larger errors more strongly.
- MAE (Mean Absolute Error) – measures average absolute prediction error.

These metrics allow quantitative comparison between the two forecasting methods.

5. ARIMA Model

5.1 Why ARIMA?

ARIMA is widely used for time series forecasting because it can model:

- Autoregressive behaviour (dependence on past values)
- Differencing to remove non-stationarity
- Moving average effects to capture past forecast errors

It is particularly suitable for economic time series where patterns depend heavily on past values.

5.2 Model Selection

The parameters of the ARIMA model are:

- **p** – number of autoregressive terms
- **d** – degree of differencing
- **q** – number of moving average terms

A grid search was performed over possible combinations of p, d, q. The best model was selected using the Akaike Information Criterion (AIC), which balances goodness of fit and model complexity.

The model with the lowest AIC was chosen as the optimal ARIMA configuration.

5.3 Model Evaluation

The selected ARIMA model was trained on the training data and used to forecast the validation period.

Performance was measured using RMSE and MAE.

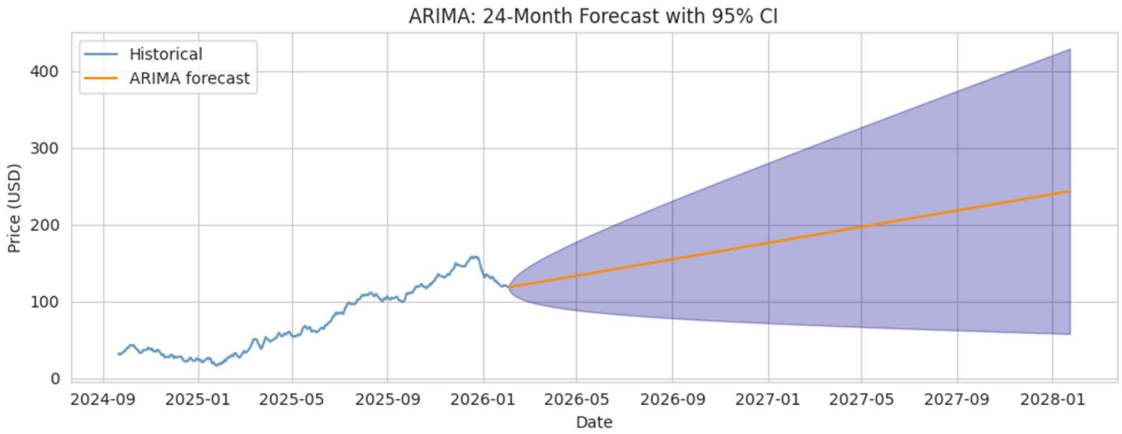

Figure 5: ARIMA validation predictions vs actual values

Residual diagnostics were also analysed.

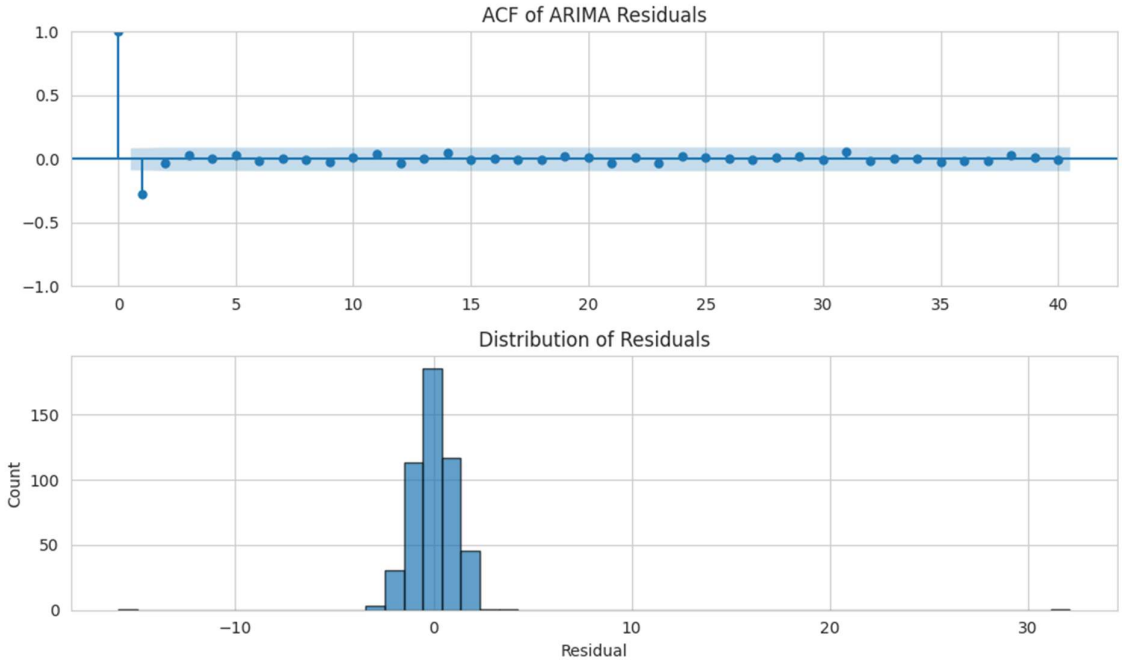

Figure 6: ACF of residuals and residual distribution

The residual ACF plot shows that most correlations fall within confidence bounds, indicating that the model captured most of the time series structure.

6. LSTM Model

6.1 Why LSTM?

While ARIMA models linear dependencies, financial time series often contain nonlinear patterns. LSTM networks are designed to model complex sequential relationships.

LSTM is particularly suitable because:

- It captures long-term dependencies

- It handles nonlinear patterns
- It performs well with large sequential datasets

These characteristics make LSTM a strong candidate for time series forecasting.

6.2 Data Preparation

Neural networks require normalized inputs. Therefore:

- The oil price series was scaled to the [0,1] range using MinMaxScaler
- A lookback window of 30 days was used

This means the model predicts the next price based on the previous 30 days of data.

The resulting training dataset shape was:

X_train: (376, 30, 1)
Y_train: (376,)

6.3 Model Architecture

The LSTM model consisted of:

- 1 LSTM layer with 32 units
- 1 Dense output layer

The model was trained using the Adam optimizer and Mean Squared Error loss function.

6.4 Model Evaluation

After training, predictions were made on the validation dataset.

Validation results:

RMSE: 5.8875
MAE: 5.0368

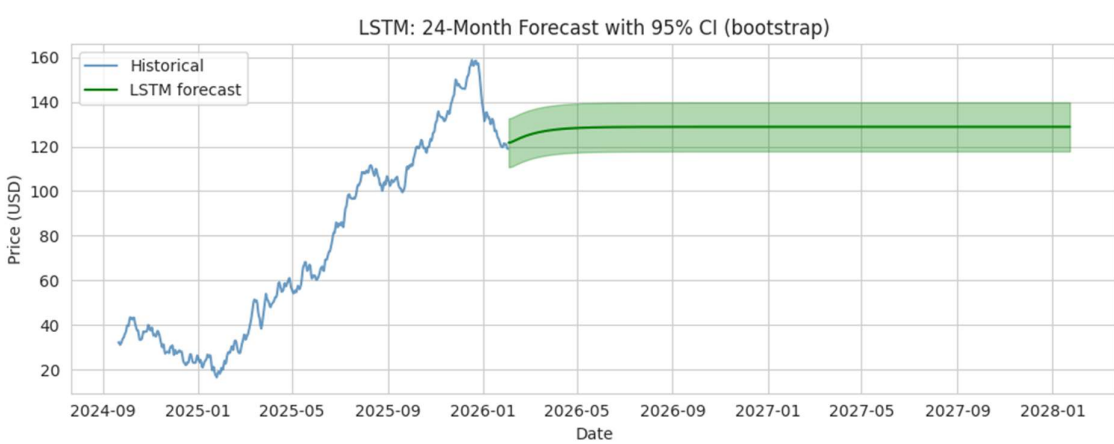

Figure 7: LSTM validation predictions vs actual values

The model was then used to generate future forecasts using recursive prediction, where each predicted value becomes input for the next prediction.

Confidence intervals were estimated using validation residuals.

7. Forecasting Results

Both models were used to forecast 24 months ahead.

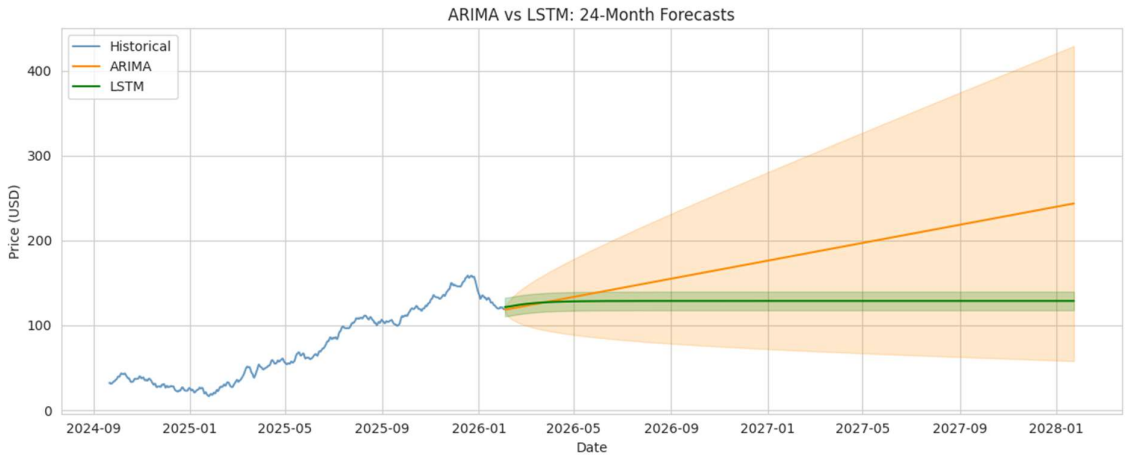

Figure 8: ARIMA vs LSTM forecast comparison

The forecast comparison illustrates differences in how each model extrapolates future price behaviour.

- ARIMA tends to produce smoother forecasts based on linear relationships.
- LSTM may capture more complex patterns but may also produce more variability.

8. Comparison of ARIMA and LSTM

Feature	ARIMA	LSTM
Model Type	Statistical	Deep Learning
Handles Nonlinearity	Limited	Strong
Data Requirements	Works with smaller datasets	Requires more data
Interpretability	High	Lower
Complexity	Low	High

ARIMA is easier to interpret and computationally efficient. However, it assumes linear relationships in the time series.

LSTM is capable of modelling nonlinear relationships and long-term dependencies but requires more computational resources and careful hyperparameter tuning.

In this analysis, the performance of the two models was compared using RMSE and MAE on the validation dataset. Overall, both models demonstrated the ability to forecast oil price movements, but their strengths differ depending on the characteristics of the time series.

9. Conclusion

This study compared ARIMA and LSTM models for oil price forecasting using daily data from 2024–2026.

Key findings include:

- The original time series was non-stationary, requiring differencing before ARIMA modelling.
- The ADF test confirmed stationarity after first differencing.
- ARIMA provided a structured statistical model with interpretable parameters.
- LSTM captured nonlinear temporal patterns using deep learning.

Both models produced reasonable forecasts, though their predictive behaviour differed. Classical models like ARIMA remain strong baselines for time series forecasting, while LSTM offers the potential to capture more complex dynamics.

Future work could improve the forecasting performance by:

- Including exogenous variables such as economic indicators
- Using hybrid ARIMA-LSTM models
- Performing hyperparameter tuning and cross-validation

10. References

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